Perceptual Image Denoising using Multi-Scale U-Net with Model Optimization

Dipanwita Rano
Texas A&M University
dipanwita22rano@tamu.edu

Dr. Duncan Walker
Texas A&M University
d-walker@tamu.edu

***Abstract*—Image denoising is a fundamental problem in computer vision, particularly for real-world noisy images captured by mobile devices. Traditional filtering methods often fail to preserve fine details while removing noise. In this work, we propose a deep learning-based approach using a U-Net architecture enhanced with perceptual loss and multi-scale training. We further explore model optimization techniques including mixed precision training and structured pruning to improve efficiency. Experimental results on the SIDD dataset demonstrate that the proposed model significantly outperforms classical filtering methods, achieving a PSNR of 38.39 dB and strong perceptual quality. Additionally, we analyze the trade-off between model compression and performance through pruning experiments.**

I. Introduction

Image denoising is a fundamental problem in computer vision, aiming to recover clean images from noisy observations. Noise in images can arise from various sources such as low-light conditions, sensor limitations, and electronic interference, especially in mobile photography. Effective denoising is critical for downstream tasks including object detection, medical imaging, and image enhancement.

Traditional image denoising techniques, such as Gaussian filtering, median filtering, and bilateral filtering, are widely used due to their simplicity and computational efficiency. However, these methods often suffer from a trade-off between noise removal and detail preservation, frequently resulting in over-smoothed outputs and loss of fine textures. Classical methods such as BM3D [1] have shown strong performance but are limited in handling complex real-world noise.

Recent advances in deep learning have significantly improved the performance of image denoising systems. Convolutional neural networks (CNNs), such as DnCNN [2], learn noise distributions directly from data and achieve superior results compared to classical methods. Encoder-decoder architectures like U-Net [3] further enhance performance by combining hierarchical feature extraction with skip connections that preserve spatial information.

In addition to pixel-wise reconstruction losses, perceptual loss functions have been introduced to improve visual quality. These losses, computed using deep feature representations, better capture human perceptual similarity compared to traditional L1 or L2 losses [4]. This is particularly important for real-world datasets where preserving texture and structural details is essential. Residual learning techniques [5] have also influenced modern denoising architectures.

Despite these advancements, two key challenges remain. First, models trained on fixed-resolution inputs often struggle to generalize across varying image sizes and noise patterns. Second, high-performing deep models are computationally expensive, limiting their deployment on resource-constrained devices.

A. Motivation

The motivation of this work is twofold. First, we aim to improve denoising performance on real-world noisy images by incorporating perceptual loss and multi-scale training. Multi-scale learning enables the model to capture noise characteristics across different spatial resolutions, leading to better generalization.

Second, we aim to address the efficiency-performance trade-off by exploring model optimization techniques such as mixed precision training and weight pruning. These techniques can significantly reduce computational cost and memory usage while maintaining acceptable performance.

B. Contributions

The main contributions of this work are as follows:

- We propose a U-Net based image denoising framework enhanced with perceptual loss for improved visual quality.
- We introduce multi-scale training to improve generalization across varying image resolutions.
- We apply mixed precision training to accelerate training and reduce memory consumption.
- We systematically analyze the impact of model pruning and demonstrate the trade-off between efficiency and performance.

C. Expected Outcomes

We expect the proposed approach to outperform traditional filtering methods and baseline CNN models in terms of both quantitative metrics (PSNR, SSIM) and perceptual quality (LPIPS). Additionally, we expect moderate pruning levels to retain most of the model performance while enabling more efficient deployment.

II. Methodology

A. Dataset

We conduct our experiments on the Smartphone Image Denoising Dataset (SIDD), which contains real-world noisy

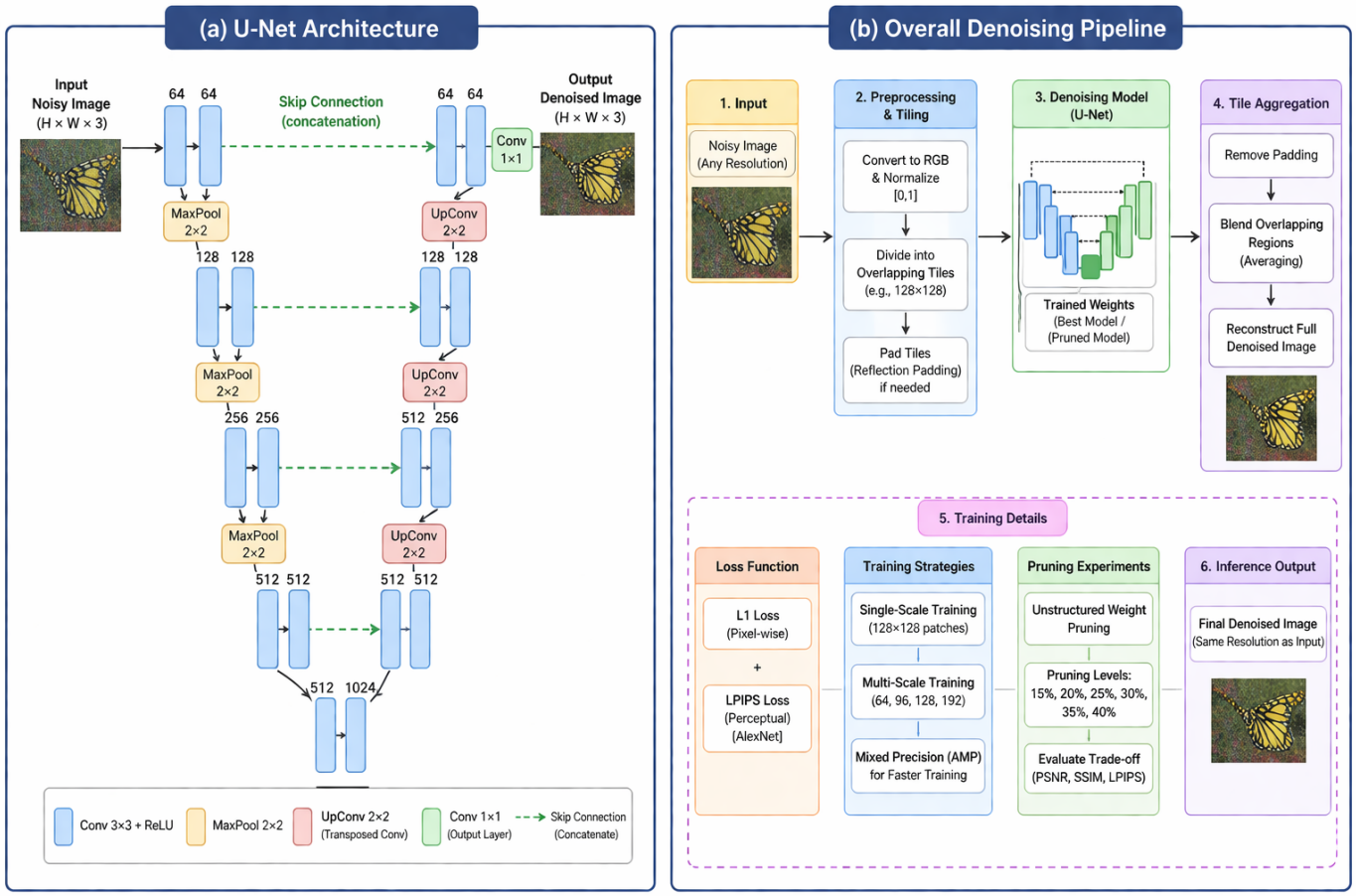

Fig. 1. Overview of the proposed image denoising framework. The left panel illustrates the U-Net architecture used for denoising, inspired by [3]. The right panel shows the complete pipeline, including preprocessing, tiling, model inference, and post-processing for reconstructing the final denoised image.

images paired with corresponding clean ground truth images. Unlike synthetic datasets, SIDD captures realistic noise patterns arising from camera sensors under varying lighting conditions. The dataset is organized into multiple scenes, each containing noisy-clean image pairs. We split the dataset into training, validation, and test sets for robust evaluation.

B. Baseline Methods

To establish a reference, we evaluate classical denoising techniques including Gaussian, median, and bilateral filtering. These traditional approaches are widely used due to their simplicity, but they often suffer from over-smoothing and loss of fine details, motivating the use of learning-based methods.

C. Network Architecture

Our model is based on the U-Net architecture [3], a widely adopted encoder-decoder framework originally proposed for biomedical image segmentation. U-Net has also been successfully applied to image restoration tasks due to its ability to capture both local and global context.

The encoder progressively downsamples the input image to extract hierarchical features, while the decoder reconstructs the clean image. Skip connections between corresponding encoder and decoder layers help preserve spatial information and fine textures, which are critical for image denoising.

Our design is further inspired by CNN-based denoising models such as DnCNN [2] and FFDNet [6], which demonstrate the effectiveness of deep convolutional architectures in modeling complex noise distributions.

D. Loss Function

We use a combination of pixel-wise L1 loss and perceptual loss. While L1 loss ensures accurate reconstruction at the pixel level, perceptual loss captures high-level semantic similarity between images. The perceptual loss is computed using deep feature representations as proposed in [4], which has been shown to correlate well with human visual perception.

E. Single-Scale Training

As an initial baseline, the model is trained using fixed-size patches (e.g., 128×128) randomly cropped from the training images. This setup allows efficient training while maintaining sufficient spatial context for learning noise patterns.

F. Multi-Scale Training

To improve generalization, we extend the training process to multiple resolutions. Specifically, patches of varying sizes

(e.g., 64×64, 96×96, 128×128, and 192×192) are used during training. For each epoch, a patch size is randomly selected, and all training samples are generated accordingly. This multi-scale strategy enables the model to learn noise characteristics across different spatial resolutions.

G. Efficient Training on GPU

Training high-resolution images requires significant computational resources. To address this, we adopt several optimization strategies:

- **Patch-based training:** Instead of full-resolution images, smaller patches are used to reduce memory consumption.
- **Mixed precision training:** Automatic mixed precision (AMP) is used to accelerate training and reduce GPU memory usage.
- **Tiled inference:** During evaluation, large images are processed in overlapping tiles to avoid memory limitations. Padding is applied to ensure compatibility with the network architecture.

H. Model Pruning

To improve computational efficiency, we apply unstructured weight pruning to the trained model. A fraction of weights with small magnitudes is removed, resulting in a sparse model. We experiment with multiple pruning levels (15% to 40%) to analyze the trade-off between model compression and denoising performance.

I. Experimental Setup

The model is trained using the Adam optimizer with a learning rate of 10^{-4}. Training is performed on a GPU using mixed precision. Performance is evaluated on the test set using standard image quality metrics. Data augmentation through random cropping further improves model robustness and prevents overfitting.

J. Evaluation Metrics

We evaluate the performance using:

- **PSNR:** Measures pixel-level reconstruction fidelity.
- **SSIM:** Captures structural similarity between images.
- **LPIPS:** Evaluates perceptual similarity using deep feature representations [4].

III. Results and Discussion

A. Comparison with Baseline Methods

We first compare our deep learning model with traditional image denoising techniques. Classical filters are computationally efficient but often fail to preserve fine details.

Method	PSNR	SSIM	LPIPS
Gaussian Filter	32.33	0.73	–
Median Filter	30.95	0.67	–
Bilateral Filter	34.36	0.82	–
U-Net (Single Scale)	**37.39**	**0.925**	0.112

TABLE I
COMPARISON WITH CLASSICAL DENOISING METHODS

The results show that the U-Net model significantly outperforms traditional filtering techniques. While classical methods tend to smooth out noise along with important details, the deep learning model preserves textures and structures more effectively.

B. Impact of Multi-Scale Training

We evaluate the effect of multi-scale training compared to single-scale training.

Model	PSNR	SSIM	LPIPS
U-Net (Single Scale)	37.39	0.925	0.112
U-Net (Multi-Scale)	**38.39**	**0.928**	**0.106**

TABLE II
EFFECT OF MULTI-SCALE TRAINING

Multi-scale training improves performance across all metrics. The model benefits from exposure to varying resolutions, enabling better generalization to diverse image sizes and noise patterns.

C. Effect of Model Pruning

To analyze the trade-off between efficiency and performance, we evaluate the model under different pruning levels.

Pruning (%)	PSNR	SSIM	LPIPS
0 (Full Model)	38.39	0.928	0.106
15	35.83	0.926	0.111
20	35.21	0.925	0.116
25	31.46	0.914	0.129
30	29.67	0.910	0.139
35	29.91	0.908	0.130
40	28.49	0.902	0.147

TABLE III
PERFORMANCE UNDER DIFFERENT PRUNING LEVELS

The results indicate that moderate pruning (15–20%) maintains acceptable performance with only a small drop in PSNR and SSIM. However, aggressive pruning beyond 25% leads to a sharp degradation in performance, indicating that the U-Net architecture is sensitive to excessive weight removal.

D. Inference Performance and Trade-offs

For inference, the multi-scale trained model (without pruning) achieves the best performance in terms of both quantitative and perceptual metrics. This model is suitable for high-quality applications where accuracy is the primary concern.

On the other hand, pruned models offer a trade-off between efficiency and performance. The 15% pruned model maintains high structural similarity and perceptual quality while reducing model complexity, making it suitable for deployment on resource-constrained devices. Higher pruning levels further reduce computational cost but result in significant degradation in output quality.

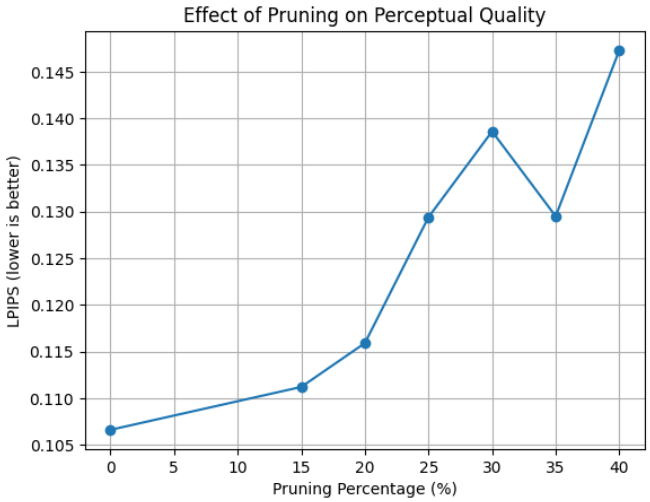

Fig. 2. LPIPS variation across different pruning levels. Lower values indicate better perceptual quality. The plot shows that perceptual quality degrades gradually with increased pruning.

E. Qualitative Results

In addition to quantitative evaluation, we visually inspect the denoised outputs produced by the proposed model. The results demonstrate that the model effectively removes noise while preserving fine textures and structural details. Compared to traditional filtering methods, which often produce over-smoothed outputs, the proposed approach maintains sharper edges and more natural image appearance.

We also observe that multi-scale training improves robustness across different image sizes, producing consistent results on both small and large images. The pruned models retain visually acceptable quality at moderate pruning levels (15–20%), although slight degradation becomes noticeable at higher pruning levels.

F. Discussion

The experimental results demonstrate that deep learning-based approaches significantly outperform traditional denoising methods. Multi-scale training further enhances performance by improving robustness across different resolutions.

The pruning experiments reveal an important insight: while moderate pruning can achieve a good balance between efficiency and performance, aggressive pruning disrupts the learned feature representations, leading to substantial quality loss. This highlights the importance of carefully selecting pruning levels for deployment scenarios.

Overall, the proposed approach achieves strong performance on real-world noisy images and provides a flexible framework that balances accuracy and efficiency.

IV. Conclusion

In this work, we presented a deep learning-based image denoising framework using a U-Net architecture enhanced with perceptual loss and multi-scale training. The proposed approach significantly outperforms traditional filtering methods, achieving a PSNR of 38.39 dB along with strong structural and perceptual similarity metrics.

Multi-scale training was shown to improve generalization across varying image resolutions, while mixed precision training enabled efficient utilization of GPU resources. Additionally, we explored model pruning as a lightweight optimization strategy and demonstrated that moderate pruning (15–20%) achieves a favorable balance between performance and efficiency.

Overall, the proposed pipeline provides a robust and efficient solution for real-world image denoising tasks, particularly for applications involving mobile imaging and resource-constrained environments. The results highlight the effectiveness of combining perceptual learning with multi-scale training for real-world denoising tasks.

V. Limitations

Despite strong performance, the proposed approach has several limitations. First, the model relies on supervised learning and requires paired noisy-clean datasets, which may not be available in practical scenarios. Second, while tiling enables inference on large images, it introduces additional computational overhead and may lead to minor boundary artifacts.

Furthermore, the U-Net architecture, although effective, can be computationally expensive for deployment on edge devices. The pruning experiments also reveal that aggressive pruning leads to significant degradation in performance, indicating sensitivity of the model to structural changes.

VI. Future Work

Future work can explore several directions to further improve the proposed approach. First, incorporating unsupervised or self-supervised learning techniques could reduce dependence on labeled datasets. Second, advanced architectures such as attention-based networks or transformer-based models may further enhance denoising performance.

Additionally, structured pruning and quantization techniques can be investigated to achieve better compression while maintaining performance. Real-time deployment optimizations, such as TensorRT or hardware-specific acceleration, can further improve inference speed.

Finally, extending the model to handle video denoising or low-light enhancement tasks presents an interesting direction for future research.

References

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